Evaluating the Impact of Congestion Pricing on Taxi Demand and Air Quality in Manhattan's CBD

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Github repo with commit

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1 Introduction

New York, specifically the Manhattan CBD, is an area accustomed to high volumes of traffic. To combat this, the Metropolitan Transportation Authority (MTA) introduced a congestion fee for most vehicles going in and out of the area, inclusive of Yellow Taxis, and High Volume For-Hire Vehicles (HVFHV). Despite having only been implemented on January 5, 2025, the federal government has already threatened to terminate this initiative. This raises the question of whether the policy is worth sustaining. Specifically, what was the impact of the January 2025 CBD congestion fee on for-hire and taxi traffic volumes, and did it correspond to measurable changes in air pollution levels in Manhattan's CBD?

In this paper, we will implement machine learning models to predict the CBD trip counts, and the air quality using two methods; linear regression and random forest regression. These measures are of note, as some of the main motivations for introducing this congestion fee was to reduce traffic and improve air quality (insert citation). This paper aims to develop a comprehensive analysis to guide the federal government and other stakeholders related to policy making into deciding whether to axe the initiative or not.

accusations that congestion pricing is a "plumber's tax" on blue-collar workers that wasn't democratically decided, a cynical cash grab by a mismanaged Metropolitan Transportation Authority (MTA), or a scheme that will flood outer-borough streets with displaced traffic. Critics have warned of economic fallout, calling the toll elitist. (from the guardian)

1.1 Datasets

A variety of datasets were used for our investigation. The main dataset used was the TLC Taxi Trip Record data and the TLC HVFHV Trip Record data from the NYC Taxi and Limousine Commission, which included trip records (datetime of pickup/drop off, fare amount, etc.) from Yellow Taxis and HVFHVs respectively. Daily estimated subway ridership was also a resource, which noted the amount of people that tapped onto the NYC subway each day. Air Quality data was used, from the NYC environment and health data portal. This dataset outlines the real-time Air Quality: PM2.5 in NYC, containing hourly measurements of PM2.5, in micrograms per cubic meter of air (µg/m3).

In these datasets, only records within the 6 month interval from November 2024 to April 2025 were

used. Since the congestion pricing began in January 5, 2025, this interval captures before and after it was implemented. The following table provides a summary of all the datasets:

Datasets	**Instances**	**No. features**
TLC Yellow Taxi Trip Record Data	22,483,319	19
TLC HVFHV Trip Record Data	121,092,373	24
Real-Time Air Quality: PM2.5 in NYC	55,272	4
MTA Subway Ridership	1,887	15

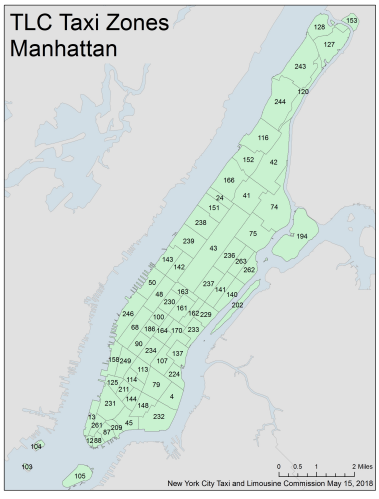
Table 1: Datasets shape

2 Preprocessing

Given that these datasets are large and varied, it was imperative that the data was preprocessed to ensure consistency before the statistical models were implemented.

2.1 Yellow Taxi and High Volume for Hire Vehicle Data

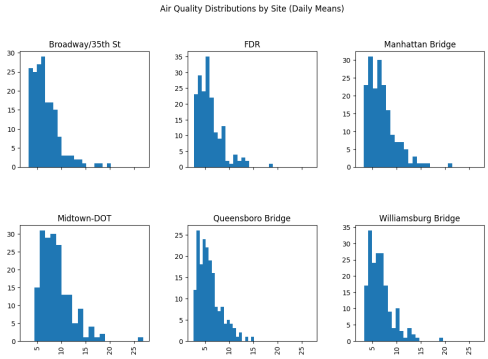
- Column names across the Yellow Taxi and HVFHV datasets were standardized to prepare them later to be joined.
- Timestamps, distances, fares, tips, tolls, and fees were converted into a consistent format.
- Basic quality rules were applied; this involved ensuring numeric values, such as passenger counts and fare amounts were non-negative. Extremely high/low values were also omitted (i.e. for distance/durations).
- Trips that were not dispatched within the Manhattan CBD area were filtered out. This is defined as trips with either a pick up or drop off location in the area defined as the Manhattan CBD.
- Between the datasets, it was important to denote the location IDs of interest to focus the investigation. The Manhattan CBD area is defined as insert citation. In our investigation, we denoted these areas as: 50, 48, 163, 230, 161, 162, 229, 233, 170, 164, 100, 186, 68, 90, 234, 107, 137, 224, 246, 158, 249, 113, 114, 125, 211, 144, 148, 232, 45, 231, 13, 261, 209, 87, 88, 12, 4, 79.



figureTLC Taxi Zones in Manhattan. Source: New York City Taxi and Limousine Commission.

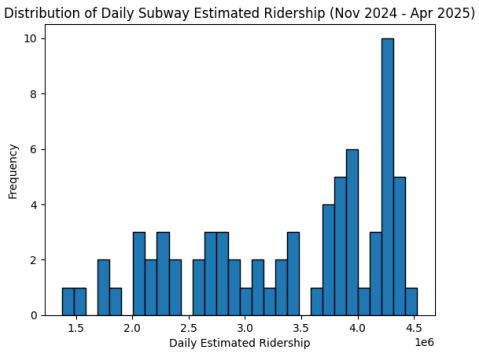
2.2 Air Quality Data

- The hourly file has a timestamp column that sometimes contains zero-width characters, RTL marks, or BOM markers that break parsing.
- For the air quality data, we restricted our analysis to monitors located within the Manhattan CBD, as identified from the official NYC Environment and Health Data Portal.
- The hourly data is then aggregated to daily means per site, tags by month
- Results are combined into one DataFrame using unionByName
- The distribution of the datasets were plotted to detail the spread, ensuring no obvious outliers.



2.3 Daily Estimated Subway Ridership

- The dates and datetimes were made consistent. The hourly file has a timestamp column that sometimes contains zero-width characters, RTL marks, or BOM markers that break parsing.
- The datasets were checked for missing values to ensure completeness.
- Since this dataset was essentially two datasets combined, (one hourly and one daily), the hourly data was aggregated to daily values in order to reduce noise and ensure a consistent level of granularity with the other datasets.
- The hourly data only had 3 features, as opposed to the 15 in the one recorded daily. Thus, the column names had to be normalized, and features that were not present in either were omitted, as it did not seem relevant to our investigation.
- Once the data had been aggregated by daily average, a histogram was plotted to visualize the spread of the data

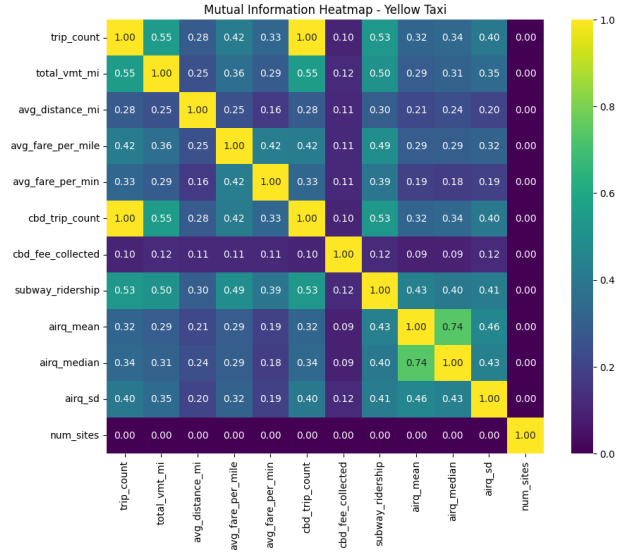


A statistical method of using $Q1, Q3 \pm 1.5 \times IQR$ was utilized to further reiterate the absence of any outliers.

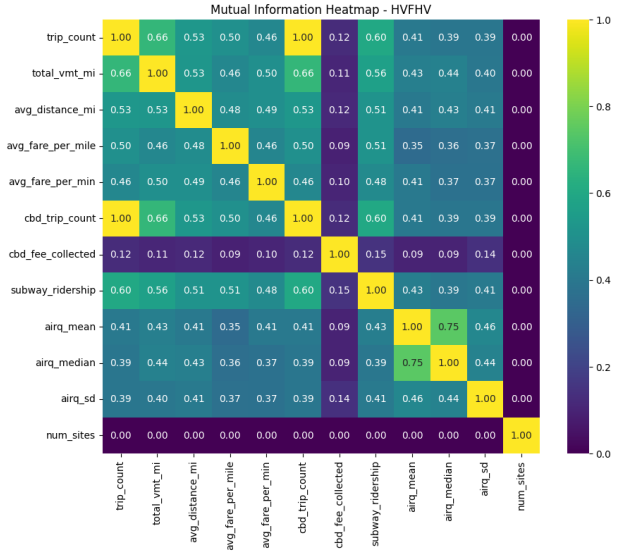
2.4 Feature Engineering and Selection

Once the data had been cleaned, features were created, such as fare per mile, fare per minute, a weekday/end flag, and post-policy indicator (whether the trip happened after the congestion pricing policy date). Trips that start or end in the CBD were also flagged.

Heatmaps were then computed, to visualize the degree of correlation. Normalized mutual information was used due to its ability to capture non-linear relationships between variables, which is useful for feature selection in the implementation of the random forest regression (insert citation). Why normalized?



figureYellow Taxi - Pearson



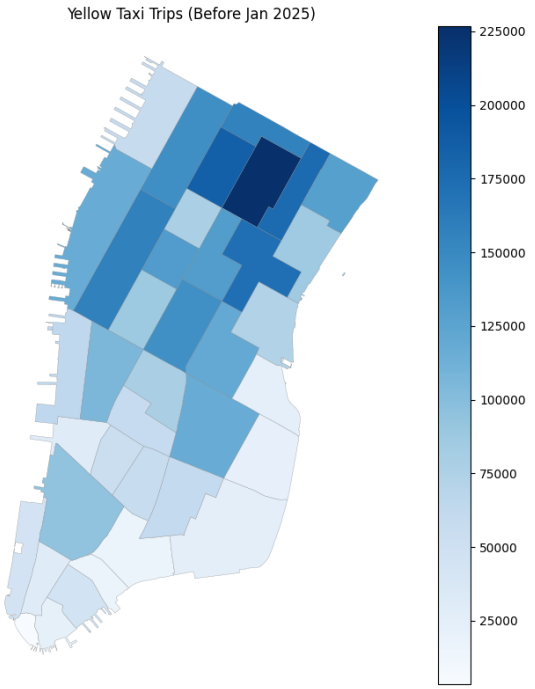
figureYellow Taxi - Mutual Information

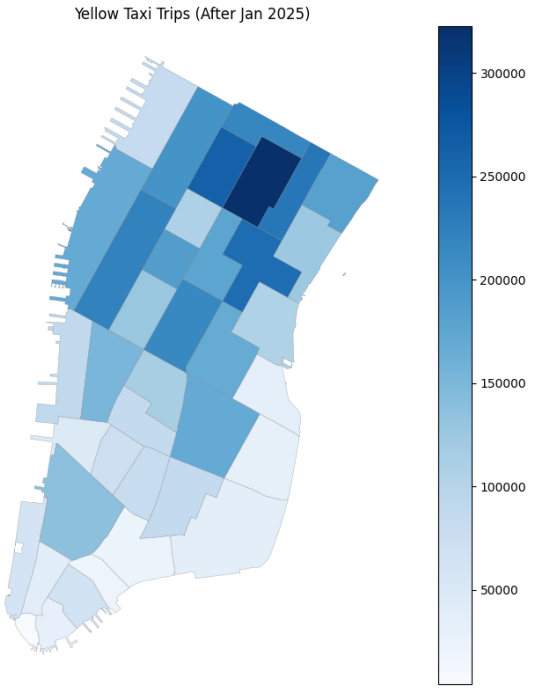
For the air quality prediction, the variables $\text{total}_v mt_m i, trip_c ount, avg_f are_p er_m ile, cbd_t rip_c ount, and subway ridership$

3 Analysis and Geospatial Visualisation

We can see yellow taxis were not as affected by congestion pricing as the number of trips dispatched before and after are relatively similar. This is supported by external reports. "Concerns that riders would abandon taxis to avoid the added $0.75 congestion fee have also proven unfounded. January was the most prosper$

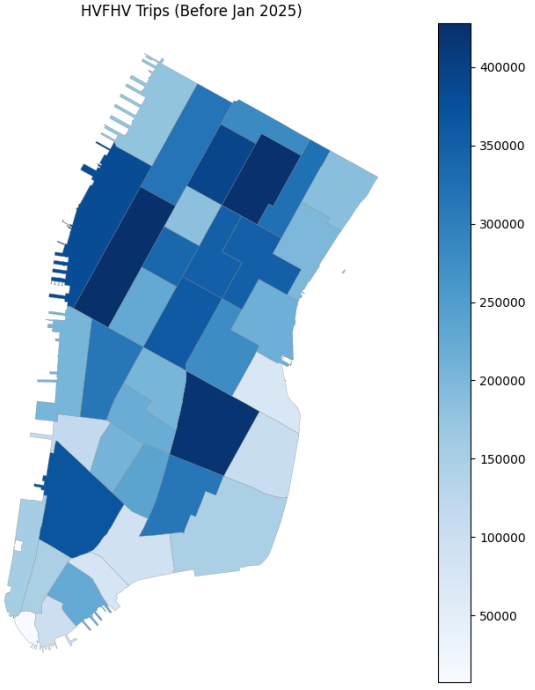
0.48

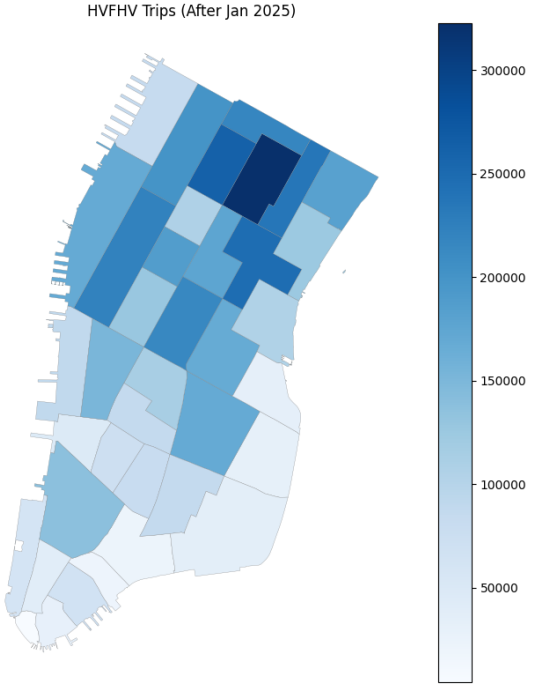




Yellow Taxi Trips (Before vs After Jan 2025)

0.48

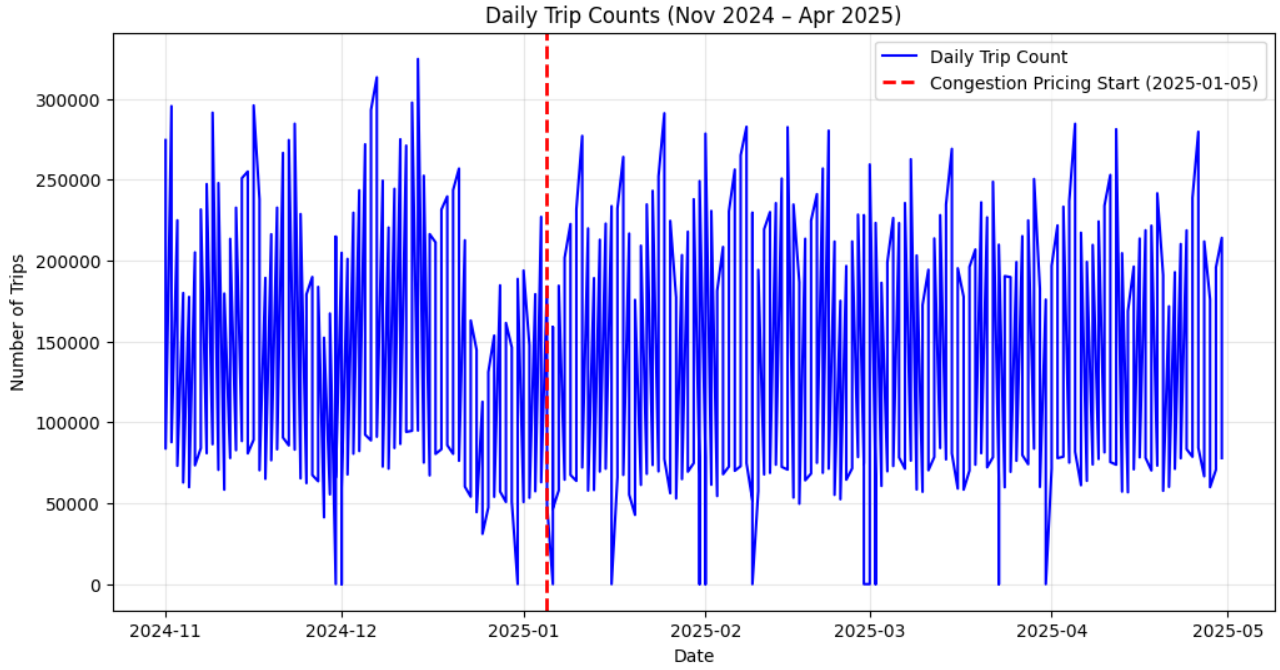




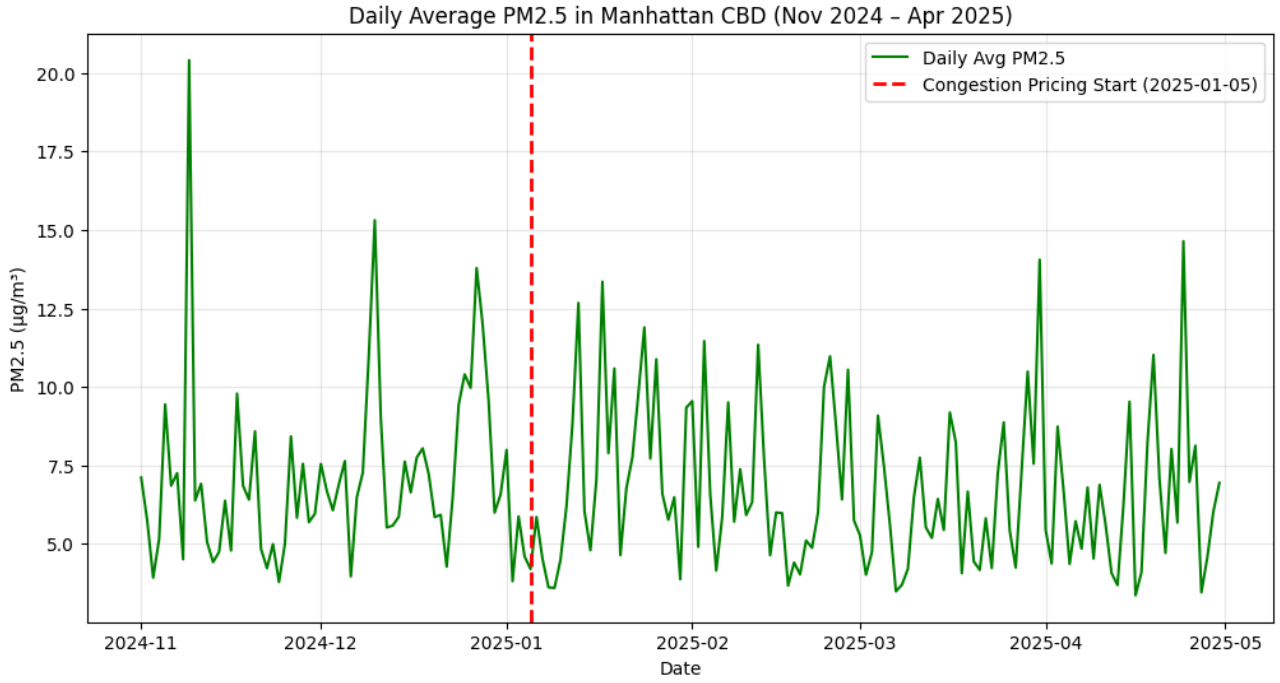
HVFHV Trips (Before vs After Jan 2025) Comparison of Yellow Taxi and HVFHV trip distributions before and after congestion pricing.

HVFHV dispatches within the CBD changed drastically before and after congestion pricing. This achieved a decrease in trips in the Manhattan CBD made by HVFHV. "Since January, an average of 70,000 fewer vehicles have entered the congestion zone each day, according to the MTA." (the guardian)

0.4



CBD Trip Trends (Before vs After Jan 2025) In the graph above, we can see that after congestion pricing was implemented, the peaks of daily trip counts were generally lower than before. This could suggest a correlation between congestion pricing, and traffic within the Manhattan CBD area. 0.4



Air Quality Trends (Before vs After Jan 2025) Trends in CBD trip counts and air quality before and after congestion pricing.

In the graph above, we can see a slight downwards trend, but it begins to spike again in the month of April. A visual observation of this graph might suggest the presence of an outlier, especially in regards to the first main spike, however, using the outlier formula, no outliers were detected in the Air Quality dataset.

4 Modelling

Linear Regression and Random Forest Regression. points to dicsuss: model implementation.

4.1 Linear Regression

First, the spark panel data was converted to a Pandas

4.2 Random Forest Regression

Motivation for using this model was to help capture the non-linear relationships in the data, as displayed in the mutual information heat maps. GridSearchCV was used to tune the hyperparameters. Features were selected based on the best mutual information scores, which were:

5 Discussion

points to discuss: model evaluation

Model	**Target**	R^2	**RMSE**	**Key Features (Top Coeffs / Importa**
LASSO Regression	Trip Count	0.98	8458	+ total_vmt_mi (9,5401) + avg_fare_per_mile (2,870) – avg_distance_mi (-15,712.90) – avg_fare_per_min (-1,483.47)
LASSO Regression	Air Quality (PM2.5 mean)	2.22e-16	2.62	– post_policy (-0.13) all other coefficients ≈ 0
Random Forest	Trip Count	0.945	16,424	total_vmt_mi (0.35) avg_fare_per_min (0.27) is_yellow (0.24)
Random Forest	Air Quality (PM2.5 mean)	-0.211	2.97	subway_ridership (0.23) trip_count (0.20) cbd_trip_count (0.20)

Table 2: Model evaluation metrics and most influential features. Positive coefficients (LASSO) and high feature importance values (RF) indicate stronger associations.

5.1 Linear Regression

isyellow and avgfarepermile had the strongest coefficients. The overall R2 was 0.88. Used LASSO, L1 regularization. When predicting for air quality, this was also repeated, but this had a really poor performance, where R2 was 0.021. The poor predictive power for air quality ($R^2 = 0.021$) suggests that factors beyond vehicle trips, such as weather or external pollution sources, are likely dominant drivers. The postpolicy coefficient had the strongest correlation.

"The strongest predictors were the service indicator (isyellow) and fare efficiency measures such as average fare per mile. This indicates that pricing dynamics and vehicle type had significant linear associations with trip counts. LASSO regularization was applied to mitigate multicollinearity and reduce overfitting, but the selected features remained largely consistent with the correlation analysis" "Even after including an interaction term and applying LASSO, the regression was unable to capture meaningful variance in PM2.5 levels. This suggests that daily air quality fluctuations are not

linearly explained by taxi and HVFHV trip volumes alone, and that external drivers such as weather, cross-borough traffic, and industrial emissions likely dominate the signal."

5.2 Random Forest Regression

In this case, air quality model prediction had really poor performance, with an R2 value of -0.211. But when the random forest model was used for predicting the cbd trip count, it performed well with and R2 of 0.95

This reflects the ability of ensemble tree methods to model interaction effects and complex relationships between service type, fares, and traffic volumes. Feature importance analysis indicated that total vehicle miles traveled and average fare per mile were among the most influential predictors.Random Forest did not improve performance for air quality prediction, where R2 = -0.211 The negative score indicates that the model performed worse than a simple mean baseline, reinforcing the conclusion that the available transport features are not sufficient to predict PM2.5 levels in the Manhattan CBD.

both models converged on the same conclusion: while congestion pricing effects are clearly measurable in trip counts, the policy's effect on air quality is far more difficult to isolate. Linear regression highlighted strong linear associations for trip counts, while Random Forest improved predictive accuracy by capturing non-linearities. For air quality, both models demonstrated that additional explanatory variables beyond taxi/HVFHVs are required

6 Recommendations

The data suggests that the taxi dispatch and HVFHV data alone is not enough to explain the air quality; there is not as much of an affect as desired, which was part of the reason for the congestion fee in the first place. However, it has correlated to less traffic, especially in the Manhattan CBD. It has had little to no effect on the livelihoods of yellow taxi drivers, but HVFHV have been greatly discouraged as a result of congestion pricing. Recommendation? continue congestion pricing, federal government should not interfere because some of the goals have been achieved. Because HVFHV trip counts fell sharply post-policy (Figure X), continuing congestion pricing may reduce overall CBD congestion while maintaining stable taxi driver earnings (Table Y).

7 Conclusion

Basically write a summary here